

Brutal Series - Paper B56 (Electric Current & its Effects | Comparing Quantities | Forests | Data Handling)

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. Four identical cells, each marked 1.5 V, are joined end to end (positive of one to negative of the next) to make a battery. A fifth cell is then added the same way. What is the total voltage of the five-cell battery?

[Electric Current & its Effects]

- (A) 5 V
- (B) 6 V
- (C) 1.5 V
- (D) 7.5 cells
- (E) 6.5 V
- (F) 9 V
- (G) 7.5 V

2. A shopkeeper buys a fan for Rs 400 and marks it up. He sells it for Rs 470. What is his profit percent on the cost price? [Comparing Quantities]

- (A) 7%
- (B) 70%
- (C) 17%
- (D) 14.9%
- (E) 18.75%
- (F) 17.5%
- (G) 20%

3. In a forest the tallest trees form a continuous green roof, below them are medium trees, then shrubs, then ground plants. A bird that needs the most sunlight and open sky will nest in which layer? [Forests]

- (A) The forest floor (herb layer)
- (B) Underground among the roots
- (C) The leaf litter
- (D) Inside a tree trunk
- (E) The shrub layer
- (F) The canopy (topmost tree layer)
- (G) The soil
- (H) The understorey

4. Five readings are 12, 15, 18, 21 and 24. Find their mean, then state by how much the largest reading exceeds the mean. [Data Handling]

- (A) Mean 18; exceeds by 12
- (B) Mean 20; exceeds by 4
- (C) Mean 18; exceeds by 3
- (D) Mean 90; exceeds by 6
- (E) Mean 18; exceeds by 0
- (F) Mean 18; exceeds by 24
- (G) Mean 15; exceeds by 9
- (H) Mean 18; largest exceeds it by 6

5. A nichrome heating wire glows when 5 A flows through it. It is replaced by a thinner nichrome wire of the same length and same material. For the same battery, what happens to the heat produced and why? [\[Electric Current & its Effects\]](#)
- (A) More heat, because thin wire has less resistance
 - (B) Less heat, because thin wire has less resistance
 - (C) No heat at all, because the wire melts instantly
 - (D) Less heat, because current cannot flow in thin wire
 - (E) More heat, because the thinner wire has greater resistance
 - (F) No change, because the length is the same
6. In a school of 250 students, 44% are girls. How many MORE boys than girls are there? [\[Comparing Quantities\]](#)
- (A) 30
 - (B) 44
 - (C) 15
 - (D) 56
 - (E) 140
 - (F) 110
 - (G) 25
7. Fallen leaves and dead animals in a forest slowly turn into a dark, crumbly material that enriches the soil. Which organisms do this, and what is the product called? [\[Forests\]](#)
- (A) Decomposers; the product is plastic
 - (B) Producers; the product is humus
 - (C) Decomposers (microbes/fungi); the product is humus
 - (D) Producers; the product is starch
 - (E) Decomposers; the product is sunlight
 - (F) Herbivores; the product is humus
8. Shoe sizes sold are 7, 3, 7, 9, 7, 3, 5. Find the mode, then the range, and add them. [\[Data Handling\]](#)
- (A) Mode 7, range 6, sum 13
 - (B) Mode 5, range 6, sum 11
 - (C) Mode 7, range 6, sum 41
 - (D) Mode 9, range 6, sum 15
 - (E) Mode 3, range 6, sum 9
 - (F) Mode 7, range 12, sum 19
9. An electric iron rated 1100 W runs on a 220 V supply. Fuses available are 3 A, 5 A, 7 A and 13 A. Which is the smallest fuse that will carry the iron's normal current without melting? [\[Electric Current & its Effects\]](#)
- (A) 7 A
 - (B) 1 A
 - (C) 11 A
 - (D) 13 A
 - (E) 3 A
 - (F) 5 A
 - (G) 6 A
 - (H) 2 A

- 10.** A jacket is marked Rs 800. A 15% discount is given, then Rs 30 is added for delivery. What does the customer finally pay? [\[Comparing Quantities\]](#)
- (A) Rs 770
 - (B) Rs 700
 - (C) Rs 685
 - (D) Rs 120
 - (E) Rs 650
 - (F) Rs 710
 - (G) Rs 740
 - (H) Rs 680
- 11.** A survey of a 200-plant forest plot finds 70% are trees and the rest are shrubs and herbs. How many plants are NOT trees? [\[Forests\]](#)
- (A) 30
 - (B) 140
 - (C) 70
 - (D) 80
 - (E) 100
 - (F) 20
 - (G) 60
- 12.** The sorted data 4, 8, 8, 10, 15 has one value added so the new median becomes 9. Which single value, inserted, makes the median exactly 9? [\[Data Handling\]](#)
- (A) The value -5
 - (B) Only values below 8
 - (C) Any value less than 4
 - (D) The value 0 only
 - (E) Only the value 8
 - (F) A value such as 1
 - (G) Any value, the median never changes
 - (H) A value such as 9 or more (e.g., 12) giving middle pair 8 and 10
- 13.** A coil of insulated wire is wound 40 times around an iron nail and connected to a cell; it lifts 8 pins. The turns are increased to 120 (same cell, same nail). Assuming lift is proportional to turns, how many pins should it now lift? [\[Electric Current & its Effects\]](#)
- (A) 24 pins
 - (B) 3 pins
 - (C) 8 pins
 - (D) 48 pins
 - (E) 12 pins
 - (F) 40 pins

14. Rs 2000 is lent at 8% simple interest per year for 2 years. What total amount (principal + interest) is returned? [\[Comparing Quantities\]](#)

- (A) Rs 2080
- (B) Rs 2240
- (C) Rs 2280
- (D) Rs 160
- (E) Rs 2400
- (F) Rs 2320

15. A simple forest food chain is: grass → deer → tiger. If hunters remove most tigers, what is the most likely SHORT-term effect on the deer and the grass? [\[Forests\]](#)

- (A) Grass increases and deer stay the same
- (B) Deer increase, then grass decreases as more deer eat it
- (C) Deer increase and grass also increases
- (D) Both deer and grass increase forever
- (E) Grass turns into trees
- (F) Deer decrease and grass increases

16. A bag has 20 red, 30 blue, 25 green and 45 yellow beads. One bead is drawn at random. What is the probability it is blue, as a fraction in simplest form? [\[Data Handling\]](#)

- (A) $\frac{1}{3}$
- (B) $\frac{30}{100}$
- (C) $\frac{2}{5}$
- (D) $\frac{30}{120}$
- (E) $\frac{1}{5}$
- (F) $\frac{1}{2}$
- (G) $\frac{3}{10}$
- (H) $\frac{1}{4}$

17. In a circuit a magnetic compass is placed under a straight wire. When the switch is closed the needle swings aside; when opened it returns. The wire is then disconnected from the cell but left lying over the compass. What will the needle do? [\[Electric Current & its Effects\]](#)

- (A) Melt the wire
- (B) Point straight up
- (C) Reverse and lock east-west
- (D) Stop working forever
- (E) Point along its usual north-south direction
- (F) Spin continuously
- (G) Swing further aside

18. A 64 cm ribbon is cut in the ratio 3 : 5. By how many centimetres is the longer piece greater than the shorter piece? [\[Comparing Quantities\]](#)

- (A) 24 cm
- (B) 2 cm
- (C) 48 cm
- (D) 16 cm
- (E) 8 cm
- (F) 40 cm
- (G) 32 cm

19. Why is a forest often called the 'green lungs' of an area and a natural air cleaner? [\[Forests\]](#)

- (A) Trees take in carbon dioxide and release oxygen, and trap dust
- (B) Trees take in oxygen and release carbon dioxide all day
- (C) Trees release carbon dioxide and trap oxygen
- (D) Trees only store water, not gases
- (E) Trees make the air hotter and dustier
- (F) Trees absorb sunlight only, no gases

20. A double bar graph shows books read over 4 weeks: boys 5,7,4,6 and girls 6,5,8,5. In how many weeks did girls read MORE than boys? [\[Data Handling\]](#)

- (A) 4 weeks
- (B) All weeks
- (C) 2.5 weeks
- (D) 0 weeks
- (E) Week 4 only
- (F) 3 weeks
- (G) 2 weeks
- (H) 1 week

21. A room has 3 identical bulbs, each 60 W, all switched on together for 5 hours. Energy is measured in watt-hours (W x h). How much energy do the three bulbs use in total? [\[Electric Current & its Effects\]](#)

- (A) 150 W-h
- (B) 1800 W-h
- (C) 450 W-h
- (D) 900 W-h
- (E) 180 W-h
- (F) 60 W-h
- (G) 720 W-h

22. A bicycle bought for Rs 600 is sold at a 12% LOSS. At what price was it sold? [\[Comparing Quantities\]](#)

- (A) Rs 480
- (B) Rs 672
- (C) Rs 528
- (D) Rs 512
- (E) Rs 588
- (F) Rs 72
- (G) Rs 612
- (H) Rs 540

23. Of 120 mm of rain on a forest, the soil and roots soak up 60% and the rest runs off. How much MORE rain is absorbed than runs off? [\[Forests\]](#)

- (A) 60 mm
- (B) 36 mm
- (C) 48 mm
- (D) 120 mm
- (E) 96 mm
- (F) 72 mm
- (G) 24 mm

24. Five test scores are 40, 40, 50, 60, 60. Find the mean and the median, then state whether they are equal. [\[Data Handling\]](#)

- (A) Mean 48, median 50, not equal
- (B) Mean 52, median 50, not equal
- (C) Mean 50, median 40, not equal
- (D) Mean 50, median 50, they are equal
- (E) Mean 250, median 50, not equal
- (F) Mean 45, median 50, equal

25. A student draws a circuit with a cell, a switch and a bulb. She wants the bulb to be controllable AND the circuit to cut off by itself if too much current flows. Which TWO components, in order, best meet both needs? [\[Electric Current & its Effects\]](#)

- (A) A switch and a fuse
- (B) A compass and a switch
- (C) A coil and a magnet
- (D) A fuse and another fuse
- (E) A switch and a second switch
- (F) Two cells
- (G) A bulb and another bulb

26. Of a class, $\frac{3}{8}$ of the students walk to school. Expressed as a percentage, what fraction does NOT walk to school? [\[Comparing Quantities\]](#)

- (A) 3.8%
- (B) 65%
- (C) 58%
- (D) 8%
- (E) 12.5%
- (F) 62.5%
- (G) 37.5%

27. A student lists: (i) trees give shade, (ii) roots bind soil, (iii) leaves add humus, (iv) trees increase soil erosion. Which one statement is WRONG? [\[Forests\]](#)

- (A) (i) trees give shade
- (B) (ii) and (iii)
- (C) (iv) trees increase soil erosion
- (D) (i) and (ii)
- (E) (ii) roots bind soil
- (F) None is wrong
- (G) (iii) leaves add humus
- (H) (iii) only

28. Six daily temperatures are 15, 18, 21, 24, 27, 30 C. What is the mean, and how many days were ABOVE the mean? [\[Data Handling\]](#)

- (A) Mean 22.5 C; 3 days above
- (B) Mean 21 C; 3 days above
- (C) Mean 22.5 C; 4 days above
- (D) Mean 22.5 C; 2 days above
- (E) Mean 24 C; 3 days above
- (F) Mean 22.5 C; 0 days above
- (G) Mean 22.5 C; 6 days above
- (H) Mean 135 C; 3 days above

29. A circuit is protected by a 5 A fuse. A faulty appliance suddenly draws 7 A. By how many amperes does the current exceed the fuse rating, and what happens to the fuse? [\[Electric Current & its Effects\]](#)

- (A) 5 A over; the fuse melts
- (B) 7 A over; the fuse melts
- (C) 2 A under; nothing happens
- (D) No excess; the fuse is fine
- (E) 2 A over; the appliance speeds up
- (F) 2 A over; the fuse glows but stays whole
- (G) 12 A over; the fuse melts
- (H) 2 A over; the fuse melts and breaks the circuit

30. A phone marked Rs 1200 gets a 20% festive discount, then a further 10% off the reduced price. What is the final price? [\[Comparing Quantities\]](#)

- (A) Rs 900
- (B) Rs 720
- (C) Rs 840
- (D) Rs 864
- (E) Rs 960
- (F) Rs 816

31. In the forest water cycle, trees draw water from the soil and release vapour from their leaves, which later forms rain. This release of water vapour from leaves is called [\[Forests\]](#)

- (A) Transpiration
- (B) Evaporation from soil
- (C) Condensation
- (D) Respiration
- (E) Pollination
- (F) Germination
- (G) Photosynthesis

- 32.** In a survey of 50 people, 30 like tea. If one person is picked at random, what is the probability they do NOT like tea, as a decimal? [\[Data Handling\]](#)
- (A) 0.5
 - (B) 0.6
 - (C) 0.7
 - (D) 0.2
 - (E) 0.8
 - (F) 0.3
 - (G) 0.04
 - (H) 0.4
- 33.** Two cells of 1.5 V each can be joined two ways: both pointing the same way (series, voltages add) or pointing opposite (voltages oppose). What is the difference between the two possible total voltages? [\[Electric Current & its Effects\]](#)
- (A) 3 V
 - (B) 0.75 V
 - (C) 4.5 V
 - (D) 0 V
 - (E) 1.5 V
 - (F) 6 V
 - (G) 2 V
 - (H) 3.5 V
- 34.** A club has 120 boys and 80 girls. What percentage of the club's members are boys? [\[Comparing Quantities\]](#)
- (A) 50%
 - (B) 75%
 - (C) 120%
 - (D) 48%
 - (E) 65%
 - (F) 40%
 - (G) 60%
- 35.** Arrange these forest dwellers by who eats whom: insect, frog, snake, leaf. Which is the correct food-chain order from producer to top consumer? [\[Forests\]](#)
- (A) Snake -> leaf -> insect -> frog
 - (B) Leaf -> insect -> frog -> snake
 - (C) Leaf -> frog -> insect -> snake
 - (D) Leaf -> insect -> snake -> frog
 - (E) Snake -> frog -> insect -> leaf
 - (F) Leaf -> snake -> frog -> insect
 - (G) Frog -> leaf -> insect -> snake

36. The mean of 3, 5, 5, 8, 9 is found. A new value 20 is added. By how much does the mean INCREASE?

[Data Handling]

- (A) By about 1.5
- (B) By about 0.83
- (C) By 0
- (D) By about 3.5
- (E) By 4
- (F) By 6
- (G) By 20
- (H) By about 2.33

37. An electric bell uses an electromagnet that pulls an iron strip to strike a gong, which breaks the circuit, lets the strip spring back, remakes the circuit, and repeats. Why does the bell keep ringing instead of striking once? [Electric Current & its Effects]

- (A) The switch is held down by a person
- (B) The circuit is made and broken again and again automatically
- (C) The current keeps increasing
- (D) Sound waves push the strip back
- (E) The cell gets stronger each time
- (F) The magnet is permanent and never lets go

38. A book's price rose from Rs 50 to Rs 65. What is the percentage increase in price? [Comparing Quantities]

- (A) 65%
- (B) 35%
- (C) 15%
- (D) 30%
- (E) 20%
- (F) 13%

39. Why does cutting down a large forest (deforestation) often lead to MORE floods and LESS rain in that region? [Forests]

- (A) More roots soak water, causing floods
- (B) Trees block rain, so removing them adds rain
- (C) Trees cause floods, so removing them stops floods
- (D) Fewer roots to soak water means more runoff, and less transpiration means less rain
- (E) Deforestation has no effect on water
- (F) Cutting trees increases transpiration
- (G) Fewer trees mean more oxygen and more rain
- (H) Soil becomes harder so it absorbs more

40. Votes are A:8, B:12, C:5, D:15. What percentage of all votes did D receive? [Data Handling]

- (A) 15%
- (B) 40%
- (C) 37.5%
- (D) 30%
- (E) 25%
- (F) 35%

- 41.** From Q6, the three 60 W bulbs used 900 W-h of energy. One 'unit' on the bill is 1000 W-h, and each unit costs Rs 8. What is the cost of running the three bulbs for those 5 hours? [\[Electric Current & its Effects\]](#)
- (A) Rs 4.50
 - (B) Rs 0.90
 - (C) Rs 900
 - (D) Rs 8
 - (E) Rs 7.20
 - (F) Rs 14.40
 - (G) Rs 6
 - (H) Rs 72
- 42.** A family spends 40% of its Rs 300 daily budget on food and 35% on rent. How many rupees are left? [\[Comparing Quantities\]](#)
- (A) Rs 75
 - (B) Rs 225
 - (C) Rs 105
 - (D) Rs 50
 - (E) Rs 60
 - (F) Rs 90
 - (G) Rs 120
- 43.** A forest patch records 24 animal species: 9 birds, 7 mammals, and the rest reptiles and insects. What fraction of the species are reptiles and insects, in simplest form? [\[Forests\]](#)
- (A) $\frac{1}{4}$
 - (B) $\frac{8}{24}$
 - (C) $\frac{1}{2}$
 - (D) $\frac{1}{3}$
 - (E) $\frac{2}{8}$
 - (F) $\frac{3}{8}$
 - (G) $\frac{2}{3}$
- 44.** Seven sorted values are 10, 20, 30, 40, 50, 60, 70. Find the median, then the mean, and state which is larger. [\[Data Handling\]](#)
- (A) Median 40, mean 280, mean larger
 - (B) Median 35, mean 40, mean larger
 - (C) Median 40, mean 40, they are equal
 - (D) Median 45, mean 40, median larger
 - (E) Median 40, mean 40, they are equal
 - (F) Median 30, mean 40, mean larger
 - (G) Median 40, mean 45, mean larger
- 45.** A circuit has a cell, a bulb and an open switch. A second identical bulb is added in series, and the switch is closed. Compared with one bulb, the two bulbs in series will glow [\[Electric Current & its Effects\]](#)
- (A) Brighter, because there are more bulbs
 - (B) Dimmer, because the same cell now drives two bulbs in a row
 - (C) Exactly the same
 - (D) One bright, one off
 - (E) Not at all, ever
 - (F) Flickering forever

- 46.** An item costs Rs 500. A shop wants a 20% profit. It instead sells at Rs 600. Is the actual profit percent more or less than planned, and by how much? [\[Comparing Quantities\]](#)
- (A) More, by 20 percentage points
 - (B) Less, by 0 (it is 25%)
 - (C) More, by 10 percentage points
 - (D) Less, by 20 percentage points
 - (E) More, by 0 percentage points (it is exactly 20%)
 - (F) More, by 100 percentage points
 - (G) Less, by 5 percentage points
 - (H) More, by 80 percentage points
- 47.** Which sequence correctly shows how dead leaves return nutrients to a growing tree? [\[Forests\]](#)
- (A) Humus → dead leaves → decomposers → oxygen
 - (B) Dead leaves → decomposers → humus in soil → roots absorb nutrients → tree grows
 - (C) Tree grows → dead leaves → oxygen → roots
 - (D) Dead leaves → sunlight → humus → tree
 - (E) Roots → dead leaves → tree → decomposers
 - (F) Decomposers → sunlight → tree → humus
 - (G) Dead leaves → roots → decomposers → humus
 - (H) Dead leaves → tree → humus → roots
- 48.** A box holds twenty identical tokens stamped with the numbers 1 through 20; one token is taken out without looking. Find the probability that its number is a multiple of 6, in simplest form. [\[Data Handling\]](#)
- (A) $\frac{1}{4}$
 - (B) $\frac{2}{20}$
 - (C) $\frac{3}{20}$
 - (D) $\frac{1}{20}$
 - (E) $\frac{3}{10}$
 - (F) $\frac{1}{6}$
 - (G) $\frac{4}{20}$
 - (H) $\frac{6}{20}$
- 49.** An electromagnet's lifting power depends on turns AND current. A coil with 50 turns and 1 A lifts 10 paper clips. To lift 40 clips using the SAME 50 turns, by what factor must the current increase (assume lift is proportional to current)? [\[Electric Current & its Effects\]](#)
- (A) 3 times
 - (B) 40 times
 - (C) 8 times
 - (D) 2 times
 - (E) 4 times
 - (F) 10 times
 - (G) 50 times
 - (H) Half

50. A discount of Rs 150 is offered on an item marked Rs 1000. What discount percentage does this represent? [\[Comparing Quantities\]](#)

- (A) 10%
- (B) 20%
- (C) 1.5%
- (D) 150%
- (E) 13%
- (F) 15%

51. A forest acts like a sponge in heavy rain. Which combination of features makes this possible? [\[Forests\]](#)

- (A) Sandy soil with no roots
- (B) Hard dry soil and no plants
- (C) Leaves that repel all water
- (D) Tall trees only, with no ground cover
- (E) Bare rock and fast slopes
- (F) Roots that block water from entering
- (G) Leaf litter and humus slow the water, and roots help it soak into the soil
- (H) Concrete paths and drains

52. Group X scored 2, 4, 6 and group Y scored 3, 3, 6. Their means are compared. Which group has the higher mean and by how much? [\[Data Handling\]](#)

- (A) Y is higher by 0.5
- (B) X is higher by 1
- (C) X is higher by 0.5
- (D) They are equal; both 12
- (E) Y is higher by 2
- (F) They are equal; both means are 4
- (G) X is higher by 2
- (H) Y is higher by 1

53. Symbols: a long line with a short thick line is a cell; a crossed circle is a bulb; a gap with a lever is a switch. In a closed loop drawn with one cell, one bulb and a switch shown OPEN, will the bulb light? [\[Electric Current & its Effects\]](#)

- (A) Yes, because switches do not matter
- (B) No, because there is only one cell
- (C) Yes, because the bulb is in the loop
- (D) No, because the bulb symbol is wrong
- (E) Yes, but only dimly
- (F) No, an open switch breaks the loop so no current flows
- (G) Yes, because the cell is connected
- (H) No, because the wire is too long

54. In a recipe the ratio of sugar to flour is 12 : 18. Written in its simplest form, what is this ratio? [\[Comparing Quantities\]](#)

- (A) 4 : 6
- (B) 2 : 1
- (C) 2 : 3
- (D) 12 : 18
- (E) 6 : 9
- (F) 1 : 2

55. Suppose each tree in a grove releases 100 units of oxygen a day. A grove has 15 trees, but 3 are cut down. How many oxygen units does the grove now release per day? [\[Forests\]](#)

- (A) 1500 units
- (B) 1000 units
- (C) 1350 units
- (D) 100 units
- (E) 300 units
- (F) 1800 units
- (G) 1200 units
- (H) 900 units

56. Data: 2, 2, 3, 4, 4, 4, 5. Find the mode and the mean, then state the difference (mode - mean). [\[Data Handling\]](#)

- (A) Mode 4, mean 3.43, difference 0.57
- (B) Mode 4, mean 3.43, difference 1.57
- (C) Mode 3, mean 3.43, difference -0.43
- (D) Mode 5, mean 3.43, difference 1.57
- (E) Mode 4, mean about 3.43, difference about 0.57
- (F) Mode 4, mean 4, difference 0
- (G) Mode 2, mean 3.43, difference -1.43
- (H) Mode 4, mean 24, difference -20

57. Three identical bulbs are connected so that unscrewing any one leaves the other two glowing. Are they in series or parallel, and why? [\[Electric Current & its Effects\]](#)

- (A) Series, because each has its own path
- (B) Series, because removing one breaks the whole circuit
- (C) Parallel, because there is only one bulb working
- (D) Parallel, because they share one single path
- (E) Neither; bulbs cannot do this
- (F) Series, because the current is the same everywhere
- (G) Parallel, because each bulb has its own separate path to the cell

58. A student scores 68 out of 80 in a test. What is the score as a percentage, and does it cross a 90% distinction line? [\[Comparing Quantities\]](#)

- (A) 80%, no
- (B) 85%, yes it crosses 90%
- (C) 68%, no
- (D) 75%, no
- (E) 90%, yes
- (F) 82%, no
- (G) 85%, no it does not reach 90%

59. In a forest, who are the 'producers', and why are they essential to every other living thing there? [\[Forests\]](#)

- (A) Tigers, because they are at the top
- (B) Insects, because they pollinate
- (C) Snakes, because they eat frogs
- (D) Decomposers, because they make humus
- (E) Fungi, because they break down leaves
- (F) Green plants, because they make food using sunlight that all consumers depend on
- (G) Soil, because it holds water
- (H) Birds, because they spread seeds

60. A bar graph shows sales of 200, 250, 150 and 400 units over four months. What is the mean monthly sale, and is any single month equal to the mean? [\[Data Handling\]](#)

- (A) Mean 250; yes, month 2 (250) equals it
- (B) Mean 225; no month equals it
- (C) Mean 250; yes month 4 equals it
- (D) Mean 200; yes month 1
- (E) Mean 1000; yes month 4
- (F) Mean 250; no month equals it
- (G) Mean 300; no month equals it

61. A 220 V line runs two irons, each rated 1100 W, at the same time. The line is protected by a 13 A fuse. Does the total current stay within the fuse rating, and by how much margin? [\[Electric Current & its Effects\]](#)

- (A) Yes; total 10 A, a margin of 3 A below the fuse
- (B) Yes; total 10 A, 23 A margin
- (C) Yes; total 5 A, 8 A margin
- (D) No; total 14 A, 1 A over
- (E) No; total 20 A, 7 A over
- (F) Yes; total 11 A, 2 A margin

62. A '10 pens for Rs 50, get 2 free' offer gives 12 pens for Rs 50. What is the effective price per pen, and what percent is saved versus the Rs 5 list price? [\[Comparing Quantities\]](#)

- (A) Rs 4.17 per pen; 83% saved
- (B) Rs 5 per pen; 0% saved
- (C) Rs 4.17 per pen; 117% saved
- (D) Rs 4.17 per pen; 50% saved
- (E) Rs 5.50 per pen; 10% saved
- (F) Rs 4 per pen; 20% saved
- (G) About Rs 4.17 per pen; about 17% saved
- (H) Rs 6 per pen; 20% saved

63. Two students argue. A: 'A forest is just a lot of trees.' B: 'A forest is a community of plants, animals, decomposers and soil that depend on each other.' Who is more correct and why? [\[Forests\]](#)

- (A) Both are equally right
- (B) B, because trees are not part of a forest
- (C) Neither, a forest is only soil
- (D) B, because a forest is an interdependent system, not just trees
- (E) A, because trees are all that matter
- (F) A, because decomposers are not alive

64. A spinner has 4 red, 6 blue and 10 green equal sectors. What is the probability the arrow does NOT land on green, in simplest form? [\[Data Handling\]](#)

- (A) $\frac{1}{3}$
- (B) $\frac{7}{10}$
- (C) $\frac{2}{5}$
- (D) $\frac{3}{10}$
- (E) $\frac{1}{5}$
- (F) $\frac{1}{2}$
- (G) $\frac{1}{4}$

65. Why is the heating element of a room heater made of nichrome rather than copper, even though copper carries current well? [\[Electric Current & its Effects\]](#)

- (A) Nichrome carries no current
- (B) Nichrome is a better conductor than copper
- (C) Copper glows brighter than nichrome
- (D) Copper melts at a higher temperature
- (E) Nichrome has high resistance and a high melting point, so it glows hot without melting
- (F) Nichrome is magnetic and copper is not
- (G) Copper is too expensive for heaters
- (H) Copper has higher resistance than nichrome

66. Rs 1500 earns simple interest at 10% per year. After how many years will the interest alone equal Rs 450? [\[Comparing Quantities\]](#)

- (A) 6 years
- (B) 2 years
- (C) 4 years
- (D) 5 years
- (E) 3.5 years
- (F) 2.5 years
- (G) 1 year
- (H) 3 years

67. Why do forest soils usually have a thick, dark top layer while bare farmland soil is often pale and thin on top? [\[Forests\]](#)

- (A) Bare soil makes its own humus faster
- (B) Forest soil has no organisms
- (C) Constant leaf fall and decomposition keep adding humus to forest soil
- (D) Forests remove humus from soil
- (E) Farm soil gets more leaf litter
- (F) Farmland has more decomposers
- (G) Forests get no sunlight, bleaching the soil
- (H) Forest soil is painted dark

68. Six values are 14, 9, 9, 9, 12, 7. Find the mean and the mode, then state which is greater and by how much. [\[Data Handling\]](#)

- (A) Mean 10, mode 14; mode greater by 4
- (B) Mean 10, mode 9; mean greater by 1
- (C) Mean 11, mode 9; mean greater by 2
- (D) Mean 10, mode 9; they are equal
- (E) Mean 60, mode 9; mean greater
- (F) Mean 10, mode 9; mode greater by 1

69. A torch uses 2 cells of 1.5 V in series and the bulb glows nicely. A child inserts a third 1.5 V cell but puts it in BACKWARDS (opposing the other two). What total voltage now drives the bulb? [\[Electric Current & its Effects\]](#)

- (A) 2.25 V
- (B) 1.5 V
- (C) 4.5 V
- (D) 7.5 V
- (E) 0 V
- (F) 6 V

70. 200 candidates take an exam and 65% pass. If 14 of those who failed reappear and all clear it next time, what percent of the original 200 have now passed in total? [\[Comparing Quantities\]](#)

- (A) 74%
- (B) 68%
- (C) 72%
- (D) 75%
- (E) 79%
- (F) 70%
- (G) 65%

71. A reserve has 40 deer and 2 tigers. Conservationists say a healthy ratio is 25 deer per tiger. Compared with the healthy ratio, is the current deer-per-tiger figure too high or too low? [\[Forests\]](#)

- (A) Too low; 2 deer per tiger
- (B) Too high; 40 deer per tiger
- (C) Too high; it is 20:... wait, $40/2 = 20$ deer per tiger, which is below 25, so too LOW
- (D) Too high; 80 deer per tiger
- (E) Too high; 42 deer per tiger
- (F) Exactly right; 20 deer per tiger
- (G) Too low; 10 deer per tiger

72. Daily attendance for a week (5 days) is 22, 24, 20, 26, 23. On how many days was attendance BELOW the mean attendance? [\[Data Handling\]](#)

- (A) 4 days
- (B) 3.4 days
- (C) 0 days
- (D) 2 days
- (E) 5 days
- (F) 1 day
- (G) 2.5 days

73. An MCB (miniature circuit breaker) is used instead of a fuse. After a fault trips it, the home owner can restore power by [\[Electric Current & its Effects\]](#)

- (A) Pouring water on it
- (B) Cutting the main wire
- (C) Connecting it backwards
- (D) Flipping the MCB switch back on once the fault is fixed
- (E) Adding more cells
- (F) Replacing a melted wire inside it
- (G) Buying a whole new MCB each time

74. A trader buys 100 eggs at Rs 8 each and sells all at Rs 9.50 each. What is the overall profit percent? [\[Comparing Quantities\]](#)

- (A) 12.5%
- (B) 1.5%
- (C) 15%
- (D) 16%
- (E) 19%
- (F) 18.75%

75. What is the correct order of layers in a dense forest, from the ground UPWARD? [\[Forests\]](#)

- (A) Herb -> floor -> shrub -> understorey -> canopy
- (B) Floor -> shrub -> canopy -> herb -> understorey
- (C) Canopy -> understorey -> shrub -> herb -> floor
- (D) Understorey -> canopy -> floor -> shrub -> herb
- (E) Forest floor -> herb layer -> shrub layer -> understorey -> canopy
- (F) Floor -> canopy -> shrub -> herb -> understorey

76. Books issued on four days are 5, 10, 15, 20. A running (cumulative) total is kept. What is the cumulative total after day 3, and what fraction of the four-day grand total is that? [\[Data Handling\]](#)

- (A) 30 issued; $\frac{1}{2}$
- (B) 50 issued; $\frac{3}{5}$
- (C) 15 issued; $\frac{1}{2}$
- (D) 20 issued; $\frac{2}{5}$
- (E) 30 issued; $\frac{2}{3}$
- (F) 45 issued; $\frac{9}{10}$
- (G) 30 issued; $\frac{3}{5}$ of the grand total
- (H) 30 issued; $\frac{3}{4}$

77. A wire carrying current is brought near a plotting compass and the needle deflects 30 degrees east. The current is then reversed in direction. The needle is now expected to [\[Electric Current & its Effects\]](#)

- (A) Point to true north exactly
- (B) Deflect further, 60 degrees east
- (C) Deflect about 30 degrees to the west
- (D) Stay at 30 degrees east
- (E) Deflect 30 degrees east then explode
- (F) Stop deflecting
- (G) Spin in circles

78. A team won 15 of its 20 matches. If it must finish the season with a 80% win rate over 25 matches, how many of the remaining 5 matches must it win? [\[Comparing Quantities\]](#)

- (A) 6
- (B) 2
- (C) 20
- (D) 5
- (E) 3
- (F) 4

79. A forest near a village is cleared for farming. Within years the village faces hotter days, dustier air and a falling water table. Which single cause best links all three problems? [\[Forests\]](#)

- (A) The farm crops absorbed all the heat
- (B) The sun moved closer
- (C) Decomposers left the soil
- (D) Too much rain from the missing trees
- (E) Animals returned to the area
- (F) The river started flowing uphill
- (G) Loss of the forest, which had cooled air, trapped dust and recharged groundwater

80. A coin tossed 50 times gives 27 heads. Based on this experiment, what is the experimental probability of TAILS, as a fraction in simplest form? [\[Data Handling\]](#)

- (A) $11/25$
- (B) $46/100$
- (C) $23/50$
- (D) $27/23$
- (E) $1/2$
- (F) $23/27$
- (G) $1/50$
- (H) $27/50$

81. In an electromagnet experiment, students record pins lifted: 20 turns -> 4 pins, 40 turns -> 8 pins, 60 turns -> 12 pins. If the pattern holds, how many turns are needed to lift 18 pins? [\[Electric Current & its Effects\]](#)

- (A) 100 turns
- (B) 60 turns
- (C) 45 turns
- (D) 90 turns
- (E) 120 turns
- (F) 72 turns

82. An item marked Rs 950 is sold for Rs 760. What discount percent was given? [\[Comparing Quantities\]](#)

- (A) 18%
- (B) 20%
- (C) 2%
- (D) 25%
- (E) 80%
- (F) 19%

83. Mushrooms and certain bacteria in a forest do NOT make their own food and are not eaten as part of the main chain, yet they are vital. What role do they play? [\[Forests\]](#)

- (A) Parasites that only harm trees
- (B) Soil that stores water
- (C) Decomposers that recycle dead matter into nutrients for plants
- (D) Top predators that hunt deer
- (E) Herbivores that graze grass
- (F) Pollinators that move pollen

84. Class A scores 60, 70, 80 and Class B scores 50, 90, 70. Their means are equal, but which class is more SPREAD OUT (larger range)? [\[Data Handling\]](#)

- (A) Class A, range 20
- (B) Class B, with range 40 versus A's range 20
- (C) Class B, range 30
- (D) Class A, range 30
- (E) Class B, range 20
- (F) Class A, range 40
- (G) Both equal, range 20
- (H) Both equal, range 40

85. Which statement correctly pairs a current effect with a real device? [\[Electric Current & its Effects\]](#)

- (A) Heating effect -> electric iron; magnetic effect -> electric bell
- (B) Heating effect -> electric bell; magnetic effect -> iron
- (C) Both devices use only the magnetic effect
- (D) Heating effect -> compass; magnetic effect -> iron
- (E) Lighting effect -> bell; heating effect -> compass
- (F) Both devices use only the heating effect

86. Rs 4000 is shared among three friends in the ratio 2 : 3 : 5. How much MORE does the friend with the largest share get than the one with the smallest? [\[Comparing Quantities\]](#)

- (A) Rs 2000
- (B) Rs 400
- (C) Rs 800
- (D) Rs 2400
- (E) Rs 1000
- (F) Rs 1200
- (G) Rs 1600

- 87.** A forest covered 300 hectares; after years of clearing only 180 hectares remain. What percentage of the original forest has been lost? [\[Forests\]](#)
- (A) 60%
 - (B) 36%
 - (C) 30%
 - (D) 20%
 - (E) 120%
 - (F) 18%
 - (G) 40%
- 88.** Seven sorted ages are 12, 12, 13, 14, 16, 17, 18. The youngest leaves. What is the median of the remaining SIX ages? [\[Data Handling\]](#)
- (A) 17
 - (B) 15
 - (C) 12
 - (D) 16.5
 - (E) 13
 - (F) 14
 - (G) 16
- 89.** Coil A (40 turns) lifts 8 pins; coil B (same current) lifts 12 pins. By what percentage is coil B's lifting power greater than coil A's? [\[Electric Current & its Effects\]](#)
- (A) 4%
 - (B) 40%
 - (C) 25%
 - (D) 33%
 - (E) 60%
 - (F) 12%
 - (G) 50%
- 90.** A toy costs Rs 250 plus Rs 50 repair before sale, and is sold for Rs 360. What is the profit percent on the TOTAL cost? [\[Comparing Quantities\]](#)
- (A) 15%
 - (B) 36%
 - (C) 44%
 - (D) 30%
 - (E) 20%
 - (F) 10%
 - (G) 25%
 - (H) 12%
- 91.** Why can a thick forest stay green and grow well for centuries without anyone adding fertiliser? [\[Forests\]](#)
- (A) Wind blows in new humus daily
 - (B) Rain brings all the fertiliser
 - (C) Trees do not need any nutrients
 - (D) Decomposers continuously recycle dead matter into humus that feeds the plants
 - (E) The soil never loses nutrients because plants give them back as plastic
 - (F) Roots make their own fertiliser from air only

92. Four numbers have a mean of 7.5 and are 3, 6, 9 and one unknown. What is the unknown number?

[Data Handling]

- (A) 12
- (B) 6
- (C) 9
- (D) 18
- (E) 7.5
- (F) 10
- (G) 15
- (H) 30

93. A fuse wire is chosen so it melts at 7 A. An appliance that normally draws 6 A develops a short and tries to draw 15 A. What is the LEAST current at which the fuse acts, and does it act here? [Electric Current & its

Effects]

- (A) It acts at 7 A; since 15 A is above 7 A, the fuse melts
- (B) It acts at 1 A; melting immediately
- (C) It acts at 15 A; so it just survives
- (D) It acts at 6 A; so it never melts
- (E) It never acts
- (F) It acts at 8 A; so it holds at 15 A
- (G) It acts at 7 A; but 15 A is below it, so it holds

94. A town's population is 8000 and grows 5% in a year. The NEXT year it falls by 5% of the new figure. Is the final population equal to, more than, or less than 8000? [Comparing Quantities]

- (A) Equal to 8000
- (B) Equal to 8400
- (C) More than 8000 (8020)
- (D) Less than 8000 (it is 7980)
- (E) More than 8000 (8100)
- (F) Less than 8000 (7600)
- (G) Less than 8000 (7920)

95. Which chain of effects best shows how forests help PREVENT floods downstream? [Forests]

- (A) Leaves drink rivers -> water vanishes -> no floods
- (B) Canopy speeds rain -> fast runoff -> floods stop
- (C) Forests have no effect on floods
- (D) Trees make extra rain -> rivers overflow -> floods worsen
- (E) Soil repels water -> runoff falls -> floods stop
- (F) Trees block all rain -> no water reaches rivers -> floods stop
- (G) Canopy and litter slow rain -> roots soak water into soil -> less and slower runoff -> rivers rise gently

96. Weekday visitors are Mon 40, Tue 55, Wed 35, Thu 50, Fri 70. On how many days were visitors ABOVE the weekly mean? [\[Data Handling\]](#)

- (A) 0 days
- (B) 2 days
- (C) 1 day
- (D) Mon and Wed
- (E) 3 days
- (F) 4 days
- (G) 2.5 days

97. Why must you NEVER touch electrical appliances with wet hands or replace a blown fuse with a thicker copper wire? [\[Electric Current & its Effects\]](#)

- (A) Both make the bulb brighter
- (B) Wet hands are too cold, and copper is too cheap
- (C) Water conducts current to your body, and thick copper will not melt so it removes the fuse's protection
- (D) Wet hands are sticky, and copper is magnetic
- (E) Wet hands clean the wire, and copper is safe
- (F) Both have no effect at all
- (G) Water blocks current, and copper melts too fast
- (H) Water cools the wire, and copper conducts poorly

98. A 20-litre fuel blend is 90% petrol and the rest ethanol. To make the blend only 80% petrol by adding pure ethanol, how many litres of ethanol must be added? [\[Comparing Quantities\]](#)

- (A) 10 litres
- (B) 2 litres
- (C) 3 litres
- (D) 4 litres
- (E) 5 litres
- (F) 2.5 litres

99. A poster says: 'Save forests to save ourselves.' Which set of reasons best supports this slogan? [\[Forests\]](#)

- (A) Forests increase pollution and floods
- (B) Forests use up all the oxygen
- (C) Forests only provide wood for furniture
- (D) Forests are useful only to animals, not people
- (E) Forests block roads and must be cleared
- (F) Forests make soil infertile
- (G) Forests have no link to rainfall
- (H) Forests give oxygen, clean air, prevent floods and erosion, and supply rain and biodiversity

100. Scores are 8, 8, 8, 9, 10, 11, 11. Find the mode, the range and the mean, then identify which two of these three are equal. [\[Data Handling\]](#)

- (A) Mode 8, range 3, mean 9; none of the three are equal
- (B) Mode 8, range 3, mean 10; none equal
- (C) Mode 11, range 3, mean 9; none equal
- (D) Mode 8, range 3, mean 9; mode and mean equal
- (E) Mode 8, range 11, mean 9; range and mean equal
- (F) Mode 8, range 3, mean 9; mode and range equal